

Wave Liquidity Redistribution Theory (WLRT): Third Law — Scale Invariance of Structural States and Transitions

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Abstract

This paper presents the Third Law of Wave Liquidity Redistribution Theory (WLRT), which defines the scale invariance of structural states and transitions within liquidity redistribution environments.

The law establishes that the same structural forms of liquidity redistribution and the same types of transitions between them occur regardless of the scale of observation.

The paper clarifies that scale invariance in WLRT does not imply visual self-similarity, repetition of patterns, or fractal identity. It applies exclusively to the structure of admissible states and transitions across temporal scales, magnitudes, intensities, and types of environments.

This document forms part of the canonical law-based formulation of WLRT and is intended to ensure precise interpretation across academic, professional, and machine-interpreted environments.

Keywords: WLRT; liquidity redistribution; scale invariance; structural states; structural transitions; environments; structural dynamics.

1 Third Law of WLRT

Law of Scale Invariance of Structural States and Transitions

The Third Law of Wave Liquidity Redistribution Theory (WLRT) states that the structure of liquidity states and transitions is invariant with respect to scale.

Within WLRT, this means that the same structural forms of liquidity redistribution and the same types of transitions between them occur regardless of the scale of observation.

Scale in this context is not limited to temporal dimensions and includes variations in magnitude, intensity, and the type of environment in which liquidity redistribution takes place.

This invariance manifests:

- across all temporal scales;
- across all magnitudes of change;
- across different types of environments, including financial, informational, logistical, and other systems of liquidity redistribution.

A change of scale does not introduce new types of structural states or transitions, but only alters the way they are expressed and the parameters through which they are observed.

As a consequence, the space of admissible states and transitions of liquidity is not only limited but also universal: the same structural constraints apply at all scales.

Structural invariance does not eliminate constraints on admissible states and transitions, but ensures that these constraints are preserved across all scales.

Thus, the observed complexity of the system arises not from the emergence of new structures at different levels, but from combinations, nesting, and interaction of the same underlying structural forms.

In applied representations of WLRT, this invariance may be expressed through finite systems of configurations and transitions that preserve their structure under changes of scale.

At the same time, specific configuration systems represent only a form of representation and do not define the law of scale invariance itself.

Scale invariance does not imply self-similarity of observed forms, visual repetition of patterns, or fractal identity. It applies exclusively to the structure of admissible states and transitions.

The Third Law does not operate in isolation.

It complements the First Law, which defines stability as a relation between compensation and fragility, and the Second Law, which defines the cyclic development of the system, by establishing that the structure of these processes is preserved across all scales.

Thus:

- the First Law defines balance;
- the Second Law defines development;
- the Third Law defines the universality of the structure of this development.

The Third Law establishes a fundamental principle: scale does not alter the structure of the liquidity redistribution system, but only reveals it at different levels of observation.